
Microeconomics I

Class 2 — Monotonicity, convexity, indifference curves

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30 September 2026

Plan for today

10:00	Sign-up sheet; three snacks
10:10	Warm-up: two relations
10:20	Quiz results: $\geq$ vs $\gg$, $\forall$ and $\exists$
10:30	Monotonicity, local non-satiation
11:00	Break
11:10	Indifference and contour sets
11:20	Convexity, the MRS
11:35	Five preferences, in pairs
11:50	Homothetic, quasilinear; summary

Practicalities

- Group sign-up sheet going round: 2–5 names per group
- Problem set 1 out Tue 6 Oct, due in class Wed 14 Oct
- Office hours: today 12:00–13:00, after class
up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Three classes, three snacks

Wed 7 Oct

?

Wed 14 Oct

?

Wed 21 Oct

?

menu: chocolate bar · crisps · nuts · biscuits · fruit · wafer

Choose all three now. Hands up: three different ones?

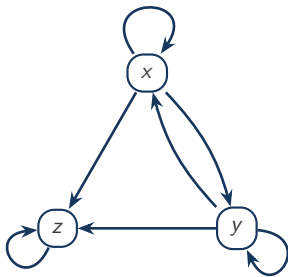
Set-up after a published experiment; source given in class.

Three snacks: what people chose

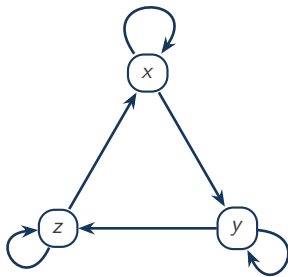
Revealed in class

Before Wednesday: choose your three, and ask yourself whether you would choose differently if asked fresh each week.

In pairs: two relations on $\{x, y, z\}$



R_1



R_2

$a \rightarrow b$ means $a R b$. Complete, reflexive, transitive, symmetric, asymmetric, antisymmetric?

Four minutes. Which one is a **rational** preference?

Revealed in class

Try it in pairs before Wednesday: the six properties for R_1 and for R_2 , and which of the two is a rational preference.

From class 1: what today builds on

$\succsim$ is **rational**: complete and transitive

MWG Def. 1.B.1

$$x \succ y \iff x \succsim y \text{ and not } y \succsim x \qquad x \sim y \iff x \succsim y \text{ and } y \succsim x$$

$\sim$ is an equivalence relation: X splits into **indifference classes**

Prop. 1.B.1

Today: what those classes look like once $\succsim$ is **monotone** and **convex**.

Diagnostic quiz: three things to fix today

Revealed in class

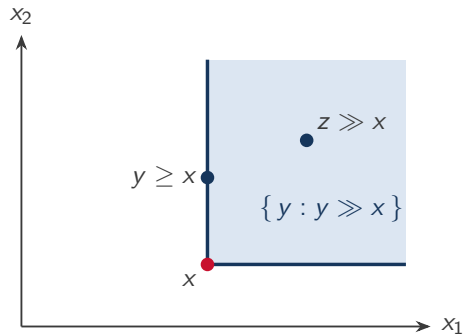
The results, and the three pieces of notation the quiz showed we need to fix.
Before Wednesday: look again at the quiz questions you were unsure about.

Two ways to say “more”

$$y \geq x \iff y_1 \geq x_1 \text{ and } y_2 \geq x_2$$

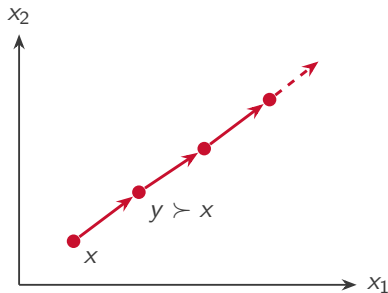
$$y \gg x \iff y_1 > x_1 \text{ and } y_2 > x_2$$

Quiz: $x = (3, 1)$, $y = (3, 2)$.



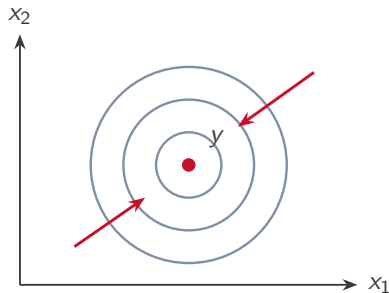
Shaded: strictly more of both. Its edges belong to $\geq$ but not to $\gg$.

Swap the quantifiers, change the claim



$$\forall x \exists y : y \succ x$$

for every bundle, a better one



$$\exists y \forall x \neq y : y \succ x$$

one bundle beats all: a bliss point

Left: no bundle is best — **non-satiation**. Its local version arrives in ten minutes.

Monotonicity

MWG 3.B

Definition: monotone, strongly monotone

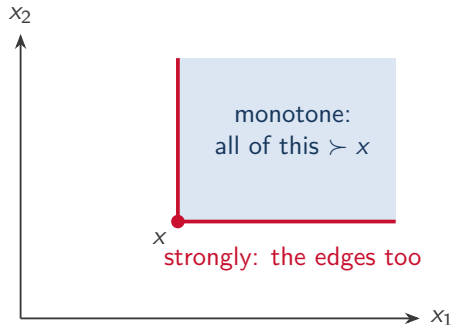
$\succsim$ on $X = \mathbb{R}_+^2$ is **monotone** if

$$y \gg x \Rightarrow y \succ x$$

strongly monotone if

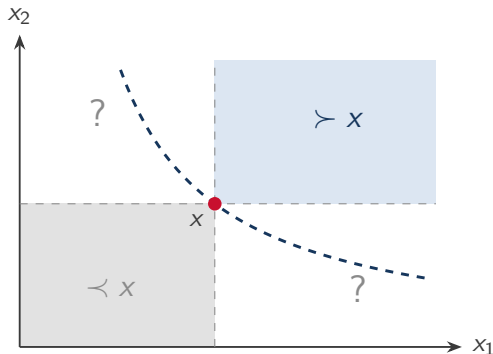
$$y \geq x \text{ and } y \neq x \Rightarrow y \succ x$$

MWG Definition 3.B.2.



Strongly monotone $\Rightarrow$ monotone. The converse fails — an example in the exercise.

Monotone: where can the indifferent bundles be?



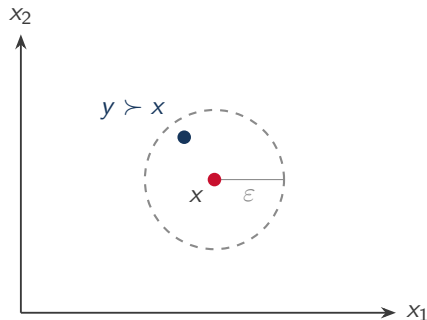
Nothing indifferent to x lies strictly north-east or strictly south-west of it. The curve through x cannot slope up.

Definition: local non-satiation

$\succsim$ is **locally non-satiated** if
for every $x \in X$ and every $\varepsilon > 0$
there is $y \in X$ with

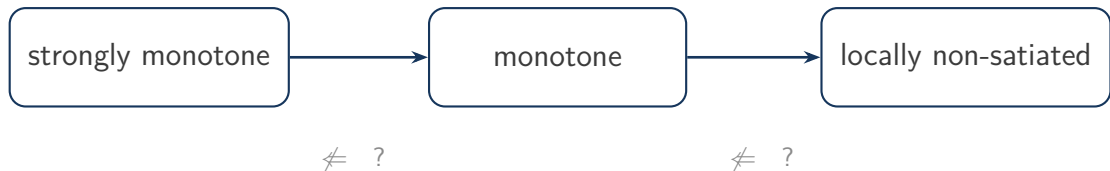
$$\|y - x\| \leq \varepsilon \quad \text{and} \quad y \succ x$$

MWG Definition 3.B.3.



$\forall x \forall \varepsilon \exists y$: the quiz's quantifier order. No direction is prescribed: y may have **less** of a good.

Three strengths of “more is better”

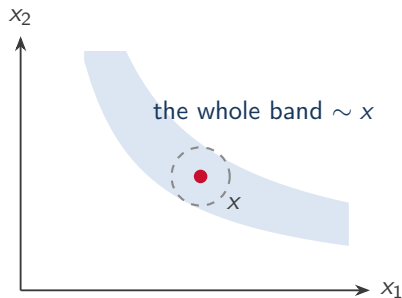


Prove monotone $\Rightarrow$ locally non-satiated in two lines. Where do you use $X = \mathbb{R}_+^2$?

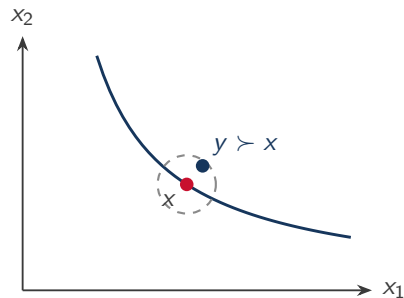
Grey: a preference with the weaker property but not the stronger one. Which of the exercise's five are they?

Revealed in class.

Local non-satiation rules out thick indifference sets



thick: fails



thin: fine

Inside a thick band, every bundle within ε of x is indifferent to x .

After MWG Figure 3.B.2, redrawn; the circle test on the right is MWG Figure 3.B.1.

In pairs: monotone? strongly? locally non-satiated?

$$x \succsim y \iff$$

monotone

strongly monotone

LNS

$$x_1 + x_2 \geq y_1 + y_2$$

$$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$$

$$x_1 - x_2 \geq y_1 - y_2$$

$$\|x - (5, 5)\| \leq \|y - (5, 5)\|$$

$$x_1 > y_1, \text{ or } x_1 = y_1 \text{ and } x_2 \geq y_2$$

Five minutes. Give a counterexample for every $\times$.

Answers revealed in class. Try it in pairs before Wednesday.

Break

Back at 11:10

Indifference and contour sets

MWG 3.B

Three sets attached to a bundle x

$$\{y \in X : y \succeq x\}$$

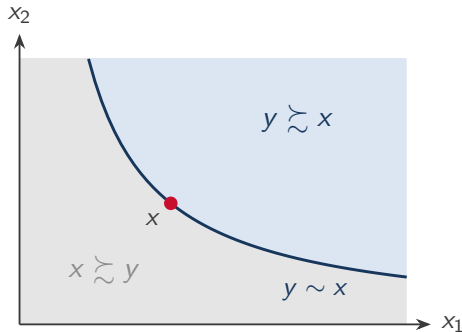
all bundles y in X such that y is at least as good as x

upper contour set $\{y \in X : y \succeq x\}$

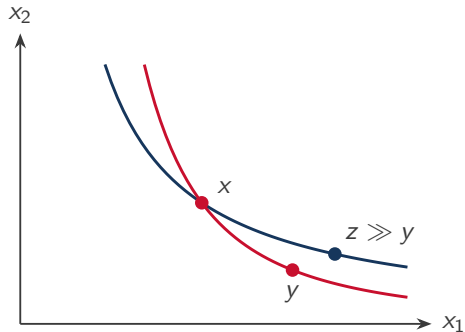
indifference set $\{y \in X : y \sim x\}$

lower contour set $\{y \in X : x \succeq y\}$

MWG 3.B, p. 43, in the text; Figure 3.B.2(b).



Indifference curves do not cross



$y \sim x$ and $x \sim z \Rightarrow y \sim z$
transitivity

$z \gg y \Rightarrow z \succ y$
monotone

Contradiction.

Rational + monotone: one indifference curve through every bundle; further out is better.

Convexity

MWG 3.B

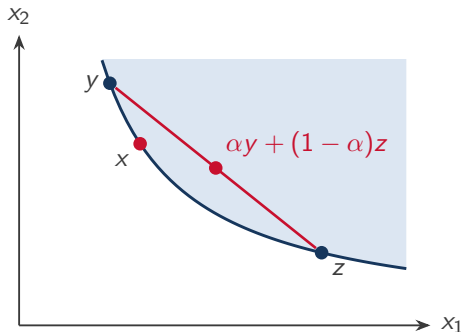
Definition: convex preferences

$\succsim$ is **convex** if every upper contour set $\{y \in X : y \succsim x\}$ is a convex set:

$$y \succsim x \text{ and } z \succsim x$$

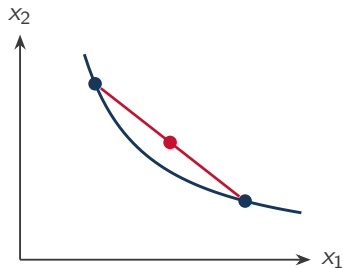
$$\Rightarrow \alpha y + (1 - \alpha)z \succsim x, \quad \alpha \in [0, 1]$$

MWG Definition 3.B.4.



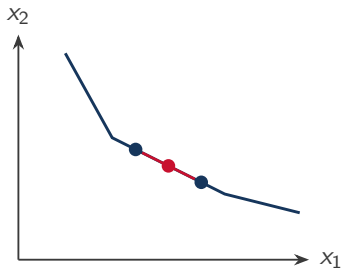
Averages are at least as good as extremes. The curve bows toward the origin.

Strictly convex, convex, not convex



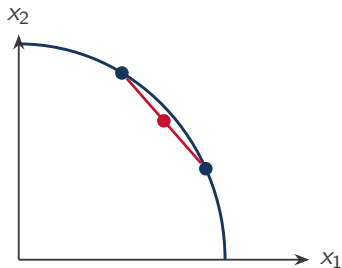
strictly convex

the mix is strictly better



convex, not strictly

a flat stretch: the mix is only $\sim$



not convex

the mix is worse

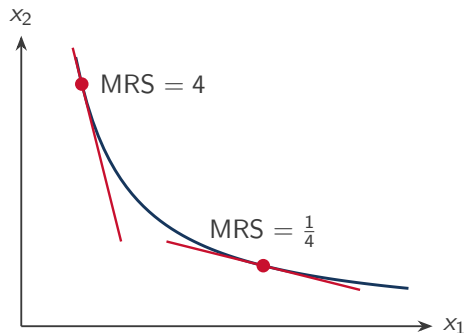
strictly convex: $y \succsim x, z \succsim x, y \neq z \Rightarrow \alpha y + (1 - \alpha)z \succ x, \alpha \in (0, 1)$

MWG Definition 3.B.5. Blue curve: an indifference set; blue dots: two bundles on it; red dot: their mix.

Revealed in class

Think beforehand: if you are indifferent between two bundles and your preferences are convex, what does that say about a mix of the two?

The marginal rate of substitution is the slope



$MRS_{12}(x)$: good 2 given up per unit of good 1, staying indifferent

$= -\text{slope of the curve at } x$

Monotone + convex: moving right along a curve, the MRS **never rises**; strictly convex: it falls.

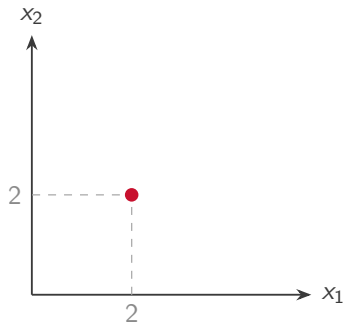
With a utility function u : $MRS_{12} = (\partial u / \partial x_1) / (\partial u / \partial x_2)$, the quiz's last question.

In pairs: five preferences on $\mathbb{R}_+^2$

	$x \succsim y \iff$	monotone	strongly	convex	strictly
perfect substitutes	$x_1 + x_2 \geq y_1 + y_2$				
perfect complements	$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$				
Cobb–Douglas type	$x_1 x_2 \geq y_1 y_2$				
quasilinear	$x_1 + \sqrt{x_2} \geq y_1 + \sqrt{y_2}$				
lexicographic	$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$				

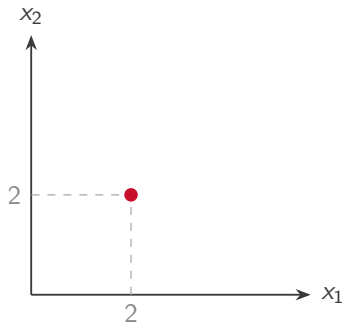
Sketch the indifference set through $(2, 2)$ for each, then fill the row. Eight minutes.

Answers: the indifference maps, I



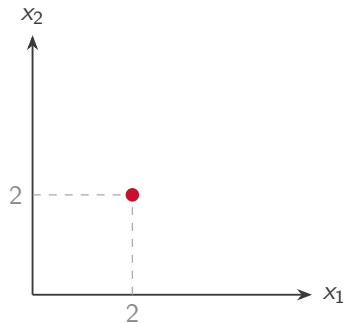
perfect substitutes

$$x_1 + x_2$$



perfect complements

$$\min\{x_1, x_2\}$$

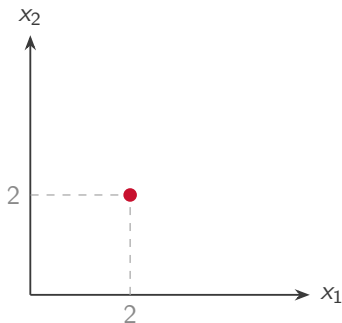


Cobb–Douglas type

$$x_1 x_2$$

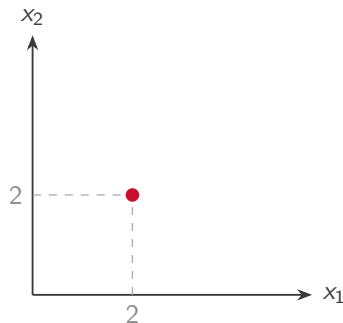
Revealed in class. Before Wednesday, sketch the indifference set through the red dot (2,2) in each panel.

Answers: the indifference maps, II



quasilinear in good 1

$$x_1 + \sqrt{x_2}$$



lexicographic

x_1 first, then x_2

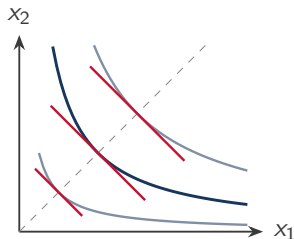
Revealed in class. Same task: the indifference set through (2,2). For the lexicographic one, also shade the upper contour set.

Answers: five preferences on $\mathbb{R}_+^2$

Revealed in class

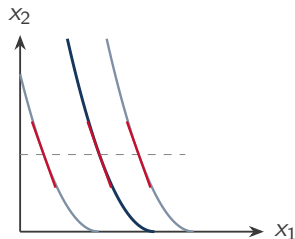
Fill the table of slide 28 in pairs before Wednesday,
with a counterexample for every cross you put.

Two families you will meet again



homothetic

same slope along a ray



quasilinear in good 1

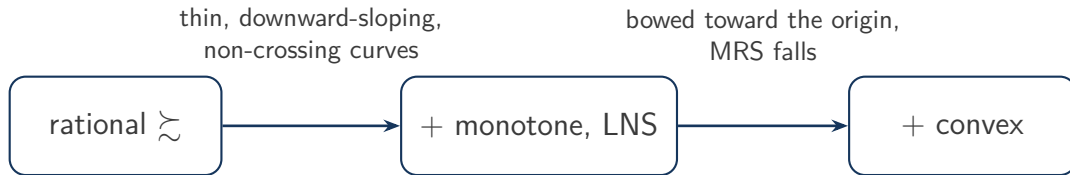
same slope along a horizontal

homothetic: $x \sim y \Rightarrow \alpha x \sim \alpha y, \alpha \geq 0$

quasilinear in good 1: $x \sim y \Rightarrow x + \alpha e_1 \sim y + \alpha e_1, \alpha \in \mathbb{R}; \quad x + \alpha e_1 \succ x, \alpha > 0$

MWG Def. 3.B.6 (monotone $\succsim$ on $\mathbb{R}_+^2$), Def. 3.B.7 (on $X = \mathbb{R} \times \mathbb{R}_+$, so $\alpha < 0$ stays in X); $e_1 = (1, 0)$.

Today in one line



Tuesday: when a number $u(x)$ can stand in for $\succsim$, and why the lexicographic one has none.
Problem set 1 goes out.

Read MWG 3.C (or Muñoz-García ch. 1) for Tuesday 6 October.